

Efficient Fine-Tuning of LLM Llama-2 Using QLoRA and Supervised Learning: A Resource-Constrained Approach

Shiven Khurdi^{*a}, Elakkiya Rajasekar^{*a},

^{*}Department of Computer Science, BITS Pilani Dubai Campus

^aEqual Contribution by said authors

1. Abstract:

In this study, we refined the LLM Llama-2 chatbot model using the Guanaco dataset through fine-tuning with the QLoRA (Quantization-based Low-Rank Adaptation) technique and Parameter-Efficient Fine-Tuning (PEFT). Managing common computational resources, we employed 4-bit precision quantization to optimize model size and memory usage. Throughout the project, we implemented QLoRA fine-tuning and PEFT, carefully managing resources during training. As a result, we developed a functional chatbot model that has been integrated into the Hugging Face model hub. This research demonstrates effective techniques for enhancing large language models under resource constraints and offers insights into efficient model training and deployment with everyday computational power.

2. Introduction:

In recent years, natural language processing (NLP) has witnessed significant advancements, particularly with the development of large-scale language models such as LLM Llama-2. These models have demonstrated remarkable capabilities in various NLP tasks, including text generation and dialogue systems. However, deploying and fine-tuning such models can be challenging, especially when confronted with resource limitations such as GPU memory constraints.

The objective of this research is to refine the LLM Llama-2 chatbot model using the Guanaco dataset through the application of the QLoRA (Quantization-based Low-Rank Adaptation) technique. The primary challenge addressed in this study is the restricted availability of GPU VRAM, with only 16 GB on a Colab T4 GPU. To tackle this limitation, we leverage 4-bit precision quantization offered by QLoRA to reduce the model's memory footprint while preserving its functionality. The development of efficient techniques for refining and deploying large language models under resource constraints is crucial for enabling widespread adoption of sophisticated NLP applications. By exploring the feasibility of fine-tuning the LLM Llama-2 model with QLoRA on limited GPU resources, we aim to contribute practical insights into optimizing model training and deployment.

This paper begins by discussing the motivation behind refining the LLM Llama-2 model and outlines the methodology employed, including dataset selection, fine-tuning and training techniques. Subsequently, the experimental results and analysis of the refined chatbot's performance are presented, followed by potential future directions of this research.

3. Literature Review:

1. The emotions and real feelings of humans are very influential on how they perceive and develop technology solutions. Modern AI technology is the pool of different ideas for generating AI-solutions that are able to understand and read both written and spoken emotions of man. The research is about explicating the functionality and applicability of sentiment analysis in an AI system that has a role of helping visitors. This chatbot is a source of very current information for those who are searching a new city for the first time. Sentiment analysis is employed to enable the chatbot to perceive and react to the mood of users, and offer responses customized according to their emotional state. Therefore, this increases user satisfaction. The LSTM + LLM system is a key aspect of this investigation that reveals the crucial elements of an advanced system which have the ability to process both the emotional state and issues of the traveler and subsequently improve the journey.
2. Machine learning and their other related AI approaches make AI an explosive topic of today. For one, the large language models or AI models whose purpose is to mimic human speech after going through amassed volumes of text ain't the exception. As is the case with ChatGPT, these models are meant to see the patterns and structures of the data and create responses in accordance with context. ChatGPT can perform different activities and give answers, just like any other LLM such as Claude v1, Google's Bard, Bison 001, Falcon, and Guanaco-65B. . This paper revolves around the basics, features, origin, proficiency, hurdles, complications, and different utilizations of LLMs.
3. The novel large language models (LLMs), like GPT-4, are referred to in the paper as being able to perform well on a variety of natural language tasks, like making a good output when writing following user directions, and producing responses that seem like a human. It is regrettable that there are very limited Vietnamese instructional datasets and pre-trained LLMs for learning, and it also takes a whole lot of time and money to train and deploy these models. The researchers approach this problem by using parameter-efficient tuning methods and open-LLMs, such as GPTJ-6B (Vietnamese) and Bloomz (Multilingual), and by exploiting instruction-following datasets from open-source projects. Evaluation of elements like helpfulness, relevancy, exactness and the level of details in the generated answers allows them to know their approach efficiency. Evaluation tasks have shown that their solution is highly relevant and even with a low-cost setup the solution can surpass the previous models.
4. Large Language Models (LLMs) have made their entry in the last few years, and they have accounted for a noticeable contribution to the development of the Natural Language Processing (NLP). This paper will look at four language models namely the BERT from Google, ELMo from the Allen Institute, GPT-3 from OpenAI as well as the LLaMA from Meta AI and discuss their constructions, train sets, evaluation performances, advantages as well as disadvantages. The paper will provide an analysis of the various models and their impact on NLP, and it will also proffer directions for future research in this field.
5. From December 2022 till April 2023 the researchers studied tweets on Large Language Models (LLM) for the development of data collecting and analyzing techniques. Further to the content of the tweets, the usernames and the emotion, they illustrated the locations of the users. The classified tweets into either positive or negative by determining the important themes and sentiment using topic modeling. In order to understand the views of the early

users they had to sort the tweets manually. This method is based on ethical data processing and lets a comprehensive analysis of the reception and issues of LLMs.

6. The study concentrates on the tuning of an LLM, Falcon, with 7 billion parameters, to mimic professorship in the "Circuit Theory" course. Pandemic classes were recorded and brought into a database during the period of pandemic. Whisper model was applied for the audio content to be correctly transcribed, as a base for Falcon 7B to be fine-tuned. Techniques like LoRA and QLoRA allowed for the training of a model on a single GPU. The comparison of a base and fine-tuned model's responses to 10 course-related questions with GPT-4 was conducted to assess the effectiveness of the prompt. Data showcase that the model has been fine-tuned successfully, thanks to which it is capable of imitating university lesson and give good answers to the course-related questions. It is this new approach that seems to be promising in education, for it can be used for interactive learning and management of online queries as well.
7. In this research, a novel chatbot that answers widely requested inquiries (FAQs) that are usually handled by people is introduced. This cutting-edge system, in contrast to conventional bots, can understand and reply to customer inquiries that are based on images in addition to words. In the event that the user is still not satisfied with the bot's reply, control is automatically passed to a human agent, who can respond with more customization and update the database accordingly. This development provides a holistic approach to FAQ interactions, combining AI and human involvement to improve user experience.
8. A chatbot functions as a virtual assistant across many platforms, such as messaging apps or smartphones. Since many businesses are incorporating chatbots to reduce operating costs, their popularity has increased. Chatbots answer consumer inquiries, freeing up human workers for more difficult work. Technological developments allow chatbots to do more than just respond text-based. In order to simplify the application process, this article suggests a chatbot option for job seekers. Through the chatbot interface, users may easily apply for jobs without having to visit the company's website or app.
9. The functionality and implications of chatbots—automated systems meant to converse with users as though they were human—are covered in this paper. In education, chatbots specifically help students by answering questions related to their coursework, helping with placement preparation, and making it easier for them to participate in extracurricular activities. By effectively supplying the needed information, this helps not only the students but also the educational departments by reducing their effort. Chatbots facilitate communication between teachers and students by providing prompt answers to questions, which eventually improves the learning process.
10. This paper presents the "College Enquiry Chatbot," an online tool intended to answer questions about education. This open-source platform, built with Rasa digital technology, utilizes Rasa Core and Rasa NLU to generate an AI Chatbot with context. According to precision, accuracy, and F1 Score criteria, the system is evaluated and the findings show promise for wider adoption because of its accessibility, correctness, and minimal maintenance requirements. This will help educational institutions and other sectors that are having query-related issues.

11. Chatbot technology is being used by educators more and more in a variety of disciplines, including teaching and learning. Technology advancement has made educational resources that are integrated with learning design, which were previously limited, abundant. Chatbots provide assurance by addressing a variety of logistical and different difficulties in education. In order to overcome educational issues, this paper offers a rule-based chatbot that was developed on Discord and shows how it may be integrated into online platforms. Preinstalled chatbot data is better understood with question-and-answer functionality.
12. The study addresses how usability and accessibility problems prevent firms from using Business Intelligence (BI) products to their full potential. Organizations use chatbots to address this, but creating BI chatbots is difficult and necessitates coordination between technical and BI specialists. This paper suggests creating tailored BI chatbots automatically using a model-driven method. The method consists of an automated component for code creation and a modeling component for designing chatbots tailored to individual users. The approach's potential for developing interactive BI chatbots customized to meet various business demands is validated by a case study.
13. This paper presents the Intelligent Career Counselling Chatbot (ICCC), a tool created to help students in the tenth and twelfth grades investigate career alternatives for postsecondary education and get their questions answered. In order for the chatbot to predict future events based on user input and present them on the GUI, users must interact with it by answering prompted questions with 1 for true and 0 for false. In addition to offering career coaching, the chatbot converses with users in general. The article lists four essential elements: machine learning techniques for prediction, adaptor and module enhancements, chatbot training, and server deployment for GUI-based interactions. This adaptable chatbot seeks to help a wide range of students discover their interests and choose a professional path.
14. An automated medical chatbot was developed for conversational engagement to provide tailored diagnosis in the context of cost-effective healthcare and improved accessibility to medical services. The system achieved a 65% recall and 71% precision in correctly detecting symptoms, achieving a 65% precision in condition detection from user inputs. The chatbot successfully offered predicted diagnoses, indicating its potential for straightforward conversational symptom analysis. This study suggests that conversational medical bots are feasible and points to the growing importance of automated medical devices in the healthcare industry.
15. The research paper looks at the way Virtual Private Servers (VPS) are used in Indonesia and emphasizes how expensive they are. It highlights how important chatbot apps are to marketing in order to efficiently convey information. The study is centered on using Python-implemented Chatbot systems with WhatsApp. As the chatbot system reads in messages, it sends pertinent information in accordance with preset criteria. But there are drawbacks as well. For example, the system can only manage 15 contacts at once, and it can't interpret messages that contain stickers, emojis, or animated gifs. Potential future improvements might include adding a function for random messages.

4. Comparative Study

Paper No.	Model Architecture	Limitations	Future Scope	Evaluation Metrics
1	LSTM + LLM	Difficulty comprehending complex contexts	Bias Mitigation and Fine-Tuning	Accuracy, Contextual Relevance
2	LLM	Computational Resources and Bias	Automation and Pattern Recognition	Accuracy
3	LLM	High costs and Dataset Availability	Dataset Expansion	GPT4 scoring method
4	Transformers, LSTM, LLMs	Contextual Understanding and Computational Costs	Development of Larger and More Complex Models and Enhanced Understanding of Natural Lang.	Accuracy, Reading Comprehension and Common-Sense Reasoning
5	LLM	Contextual Ambiguity, Negation Handling, Sarcasm and Irony	AI-Powered Education, Healthcare Innovations, Ethical Frameworks	Accuracy
6	LLM (Fine-tuned)	Scalability and Domain Specificity	Model Optimization and Efficiency, Multimodal Learning, etc	Average human-scored rating (on a scale of 1 to 7)
7	NLP+CNN+Feedback Mechanisms	Reliability, Data and Training, Usability	Enhanced Multimodal Capabilities	Accuracy
8	NLP+CNN+Feedback Mechanisms	Limited Context Understanding, Handling of Negative Sentiments	Contextual Understanding, Bias Mitigation, Scalability	Accuracy
9	LLM	Integration of Empathy, Off-Script Questions	Integration of Speech Recognition Technology, Sentiment Analysis	Accuracy
10	Rasa NLU + Rasa Core	Dependency on Data and Training, Complex Queries	Improved Training Data, Enhanced NLP Capabilities	Confusion Matrix, Precision, Accuracy, F1 Score, Confidence
11	Text Matching and Information Retrieval	Scarcity of Available Resources, Current State of NLP	Expansion of Content Coverage, Improvement in NLP	Accuracy

12	DSL for chatbot modeling	Scope of DSL, NLP Integration	Expansion of Chatbot Capabilities, Dashboard Interface Enhancement	Accuracy
13	Combination of ML algorithms + Chatterbot	Narrow Career Focus, Scope Limited to Specific Educational Levels	Diversification of Career Suggestions, Integration of Advanced ML Models	Accuracy
14	NLP + Knowledge Base	Dependency on External Recognition Engine	Implement image recognition capabilities for diagnostics, Incorporate sentiment analysis	Accuracy, Precision, Recall
15	Python + Selenium	Contact Limitation, Message Processing, Server Dependency	Random Message Feature, Hardware Exploration	Accuracy

5. Methodology

A. Dataset

Dataset chosen for this project was the guanaco dataset. Guanaco is a high-quality dataset that has been used to fine-tune state-of-the-art LLMs in the past. It contains about 10,000 rows of data. The entire Guanaco dataset is available on Hugging Face and it has the potential to achieve even greater performance on a variety of natural language tasks. This dataset is a subset of the Open Assistant dataset which is a human-generated, human-annotated assistant-style conversation corpus consisting of 161,443 messages in 35 different languages, annotated with 461,292 quality ratings, resulting in over 10,000 fully annotated conversation trees. The corpus is a product of a worldwide crowd-sourcing effort involving over 13,500 volunteers. This dataset contains message trees. Each message tree has an initial prompt message as the root node, which can have multiple child messages as replies, and these child messages can have multiple replies.

B. Model Architecture

A Llama-2 model with 7 billion parameters was selected for this study. It is built upon a state-of-the-art transformer-based architecture, leveraging principles similar to those of prominent models such as GPT-3 and BERT. It comprises multiple layers of transformer blocks, each containing self-attention mechanisms for capturing contextual dependencies within input sequences. The model's architecture includes token embeddings to represent input tokens and positional encodings to convey sequence order information. LLM Llama-2 is characterized by its large number of parameters, allowing it to capture intricate patterns and nuances in natural language data. The architecture is designed to excel in a range of NLP tasks, including text generation, sentiment analysis, and question answering, making it a versatile and powerful framework for fine-tuning and deploying advanced chatbot applications.

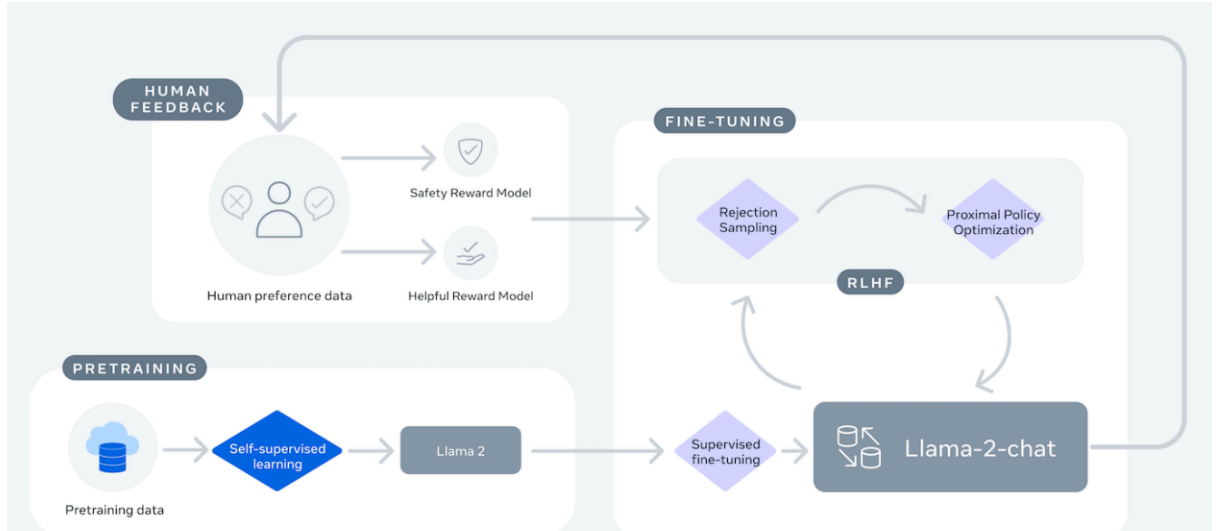


Figure 1. Llama-2 [17]

C. Fine-tuning and Training

QLoRA (Quantization-based Low-Rank Adaptation) is a novel fine-tuning technique designed for optimizing large language models under resource constraints, such as limited GPU memory. This technique combines quantization, which reduces the precision of model weights, with low-rank approximation to further compress the model's parameters. By quantizing the weights to lower bit precision (e.g., 4-bit), QLoRA significantly reduces the memory footprint of the model without compromising its functionality. Additionally, low-rank approximation is applied to factorize weight matrices into smaller matrices, further reducing the number of parameters and computational complexity during inference.

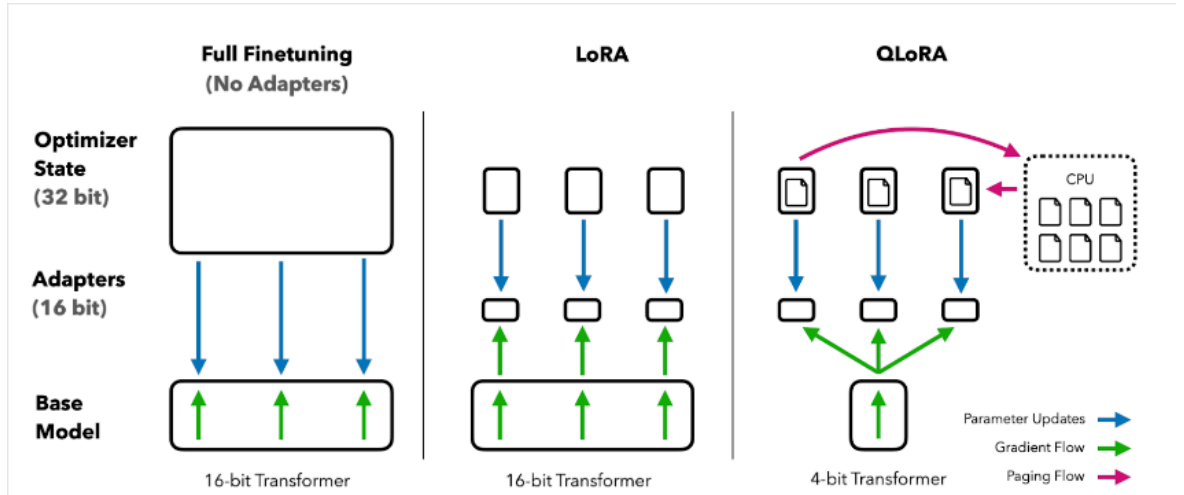


Figure 2. QLoRA [16]

In our study, we implemented the QLoRA (Quantization-based Low-Rank Adaptation) technique using a 4-bit quantization strategy with NF4 type configuration, facilitated by the BitsAndBytes library. This configuration specifies the use of 4-bit precision for quantizing the model weights, effectively reducing the memory requirements while maintaining

computational efficiency. The NF4 type configuration refers to a specific quantization scheme that optimizes the trade-off between precision and compression, tailored for fine-tuning large language models under resource constraints. By leveraging BitsAndBytes for 4-bit quantization, we achieved significant compression of the model parameters, enabling efficient utilization of the limited GPU VRAM available on the Colab T4 GPU.

Supervised fine-tuning (SFT) plays a critical role in reinforcement learning from human feedback (RLHF). The TRL library from HuggingFace simplifies the process of creating and training SFT models on custom datasets with minimal code. This library is designed for training language models using a reinforcement learning framework, starting with supervised fine-tuning and progressing through reward modeling to proximal policy optimization (PPO).

In our project, we utilized the TRL library's SFT trainer to perform supervised fine-tuning of language models on our specific dataset, laying the groundwork for subsequent reinforcement learning techniques in language model training.

Conventional fine-tuning of pretrained language models (PLMs) entails updating all parameters, a computationally demanding process necessitating extensive data. We implemented Parameter-Efficient Fine-Tuning (PEFT) that operates by solely updating a select subset of the model's most impactful parameters, significantly enhancing efficiency.

Once the model training is complete, we will save the model adapter and tokenizers for future use. Additionally, leveraging a similar API, we can seamlessly upload the trained model to Hugging Face.

6. Evaluation & Results

In this project, we leveraged our fine-tuned Llama-2-7b model, specifically designed for conversational tasks. Utilizing the Hugging Face 'transformers' library, we loaded our fine-tuned model along with its tokenizer. With a given prompt, such as "what is photosynthesis," our model generated multiple text samples aimed at answering or providing information about the prompt. These generated texts were then compared to a set of manually created reference texts, serving as benchmarks for correctness and quality. Leveraging the NLTK library, we calculated the BLEU score, a widely used metric in natural language processing tasks, to evaluate the similarity between the generated text and reference texts. With a BLEU score of 0.5, we achieved a notable level of similarity between the generated and reference texts, indicating the effectiveness of our fine-tuned Llama-2 model in generating relevant and coherent responses to the given prompt. The BLEU score was found in the range of 0.5 to 0.7 depending on the prompt and the reference provided.

Evaluation	Original Model	Fine-Tuned Model
BLEU Score on “What is Photosynthesis”	0.42	0.51

7. References

1. O. Balamurali, A. M. Abhishek Sai, M. Karthikeya and S. Anand, "Sentiment Analysis for Better User Experience in Tourism Chatbot using LSTM and LLM," *2023 9th International Conference on Signal Processing and Communication (ICSC)*, NOIDA, India, 2023, pp. 456-462
2. S. K. Routray, A. Javali, K. P. Sharmila, M. K. Jha, M. Pappa and M. Singh, "Large Language Models (LLMs): Hypes and Realities," *2023 International Conference on Computer Science and Emerging Technologies (CSET)*, Bangalore, India, 2023, pp. 1-6
3. V. -T. Doan, Q. -T. Truong, D. -V. Nguyen, V. -T. Nguyen and T. -N. N. Luu, "Efficient Finetuning Large Language Models For Vietnamese Chatbot," *2023 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*, Quy Nhon, Vietnam, 2023, pp. 1-6
4. N. T. K. Le, N. Hadiprodjo, H. El-Alfy, A. Kerimzhanov and A. Teshebaev, "The Recent Large Language Models in NLP," *2023 22nd International Symposium on Communications and Information Technologies (ISCIT)*, Sydney, Australia, 2023, pp. 1-6
5. N. P. Kumar, K. Srinivasan and D. Ramesh, "Analyzing Public Sentiment Towards LLM: A Twitter-Based Sentiment Analysis," *2023 International Conference on the Confluence of Advancements in Robotics, Vision and Interdisciplinary Technology Management (IC-RVITM)*, Bangalore, India, 2023, pp. 1-8
6. G. O. Leiva, G. H. Vigneau and D. Y. Sepúlveda, "Imitating Teaching: An Automated Approach Using Large Language Model," *2023 IEEE CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*, Valdivia, Chile, 2023, pp. 1-5
7. R. Vannala, S. B. Swathi and Y. Puranam, "AI Chatbot For Answering FAQ's," *2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT)*, Hyderabad, India, 2022, pp. 1-5
8. V. Dutt, S. M. Sasubilli and A. E. Yerrapati, "Dynamic Information Retrieval With Chatbots: A Review of Artificial Intelligence Methodology," *2020 4th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, Coimbatore, India, 2020, pp. 1299-1303
9. S. Kesarwani, Titiksha and S. Juneja, "Student Chatbot System: A Review on Educational Chatbot," *2023 7th International Conference on Trends in Electronics and Informatics (ICOEI)*, Tirunelveli, India, 2023, pp. 1578-1583
10. S. Meshram, N. Naik, M. VR, T. More and S. Kharche, "College Enquiry Chatbot using Rasa Framework," *2021 Asian Conference on Innovation in Technology (ASIANCON)*, PUNE, India, 2021, pp. 1-8
11. M. G. C. P, A. Srivastava, S. Chakraborty, A. Ghosh and H. Raj, "Development of Information Technology Telecom Chatbot: An Artificial Intelligence and Machine Learning Approach," *2021 2nd International Conference on Intelligent Engineering and Management (ICIEM)*, London, United Kingdom, 2021, pp. 216-221

12. M. Banisharif, A. Mazlounzadeh, M. Sharbaf and B. Zamani, "Automatic Generation of Business Intelligence Chatbot for Organizations," 2022 27th International Computer Conference, Computer Society of Iran (CSICC), Tehran, Iran, Islamic Republic of, 2022, pp. 1-5
13. R. Goyal, N. Chaudhary and M. Singh, "Machine Learning based Intelligent Career Counselling Chatbot (ICCC)," 2023 International Conference on Computer Communication and Informatics (ICCCI), Coimbatore, India, 2023, pp. 1-8
14. P. Srivastava and N. Singh, "Automatized Medical Chatbot (Medibot)," 2020 International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC), Mathura, India, 2020, pp. 351-354
15. A. Ramaditiya, S. Rahmatia, A. Munawar and O. N. Samijayani, "Implementation Chatbot Whatsapp using Python Programming for Broadcast and Reply Message Automatically," 2021 International Symposium on Electronics and Smart Devices (ISESD), Bandung, Indonesia, 2021, pp. 1-4
16. <https://arxiv.org/pdf/2305.14314.pdf>
17. <https://llama.meta.com/llama2/>